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SOME METAMATHEMATICAL RESULTS IN RECURSIVE ARITHMETIC

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Gödel's second incompleteness theorem, given in [2], and Löb's extension of it, given in [3], are proved in systems of arithmetic including the predicate calculus, in this note we shall give proofs for the corresponding results in recursive arithmetics[1]). The proof of the former result given in [4] is incorrect, theorem 16 of that paper is false as it stands; the proofs given here follow closely those presented in [2] and [3]. We wish to thank L. J. Pozsgay for pointing out one of the errors of [4] and M. H. Löb for helpful comments. Notation: If A is a term or equation then $[A]$ denotes the (Gödel) number of A, $se(x, y, z)$ is the number of the equation which results from equation number z when the term number x is substituted for the $(y+1)$th variable, $se(x, z) = se(x, 0, z)$ where a is the first variable, $th(y)$ is the number of the yth theorem, $nu(x)$ is the number of the xth numeral, and if t is the number of the equation $A = B$ then $e(t)$ is the number of $A \neq B$; for this and other unexplained notation see [4].

Lemma 1. *If $A_1, \ldots, A_n \vdash B$ in a system T of recursive arithmetic then there is a primitive recursive function $V(t_1, \ldots, t_n)$, whose construction depends on the given derivation, with the property*

$$th(t_1) = [A_1] \mathbin{\&} \cdots \mathbin{\&} th(t_n) = [A_n] \to th(V(t_1, \ldots, t_n)) = [B].$$

Proof. Suppose B follows from A by the substitution of term number x for the $(y+1)$th variable of A, that is

$$[B] = se(x, y, [A])$$

then, using the definition of $th(t)$,

$$\begin{aligned} th(t) = [A] &\to [B] = se(x, y, th(t)) \\ &\to [B] = th(6w(x, w(y, t)) + 1), \end{aligned}$$

hence for this derivation $V(t) = 6w(x, w(y, t)) + 1$. It is clear that, using very similar arguments, a primitive recursive function $V(t_1, \ldots, t_n)$ can be constructed for any derivation of B from $A_1, \ldots, A_n$.

Let $\{f(x) = y\}$, a function of x and y, be defined so that if m and n are numerals $\{f(n) = m\} = [f(n) = m]$. In the notation of [4]

$$\{f(x) = y\} = se(nu(x), 0, se(nu(y), 1, [f(a) = b])),$$

[1]) The lowest system in the hierarchy of recursive arithmetics, the $\mathscr{E}^n$-arithmetics given in [1], in which the proofs of this note can be carried out is $\mathscr{E}^3$-arithmetic, that is Elementary arithmetic; see [1] for further details.

here a is the first variable and b is the second; $\{f(x) = y\}$ is the number of the equation $f(a) = b$ when the xth numeral is substituted for a and the yth numeral for b.

Lemma 2. *For each function $f(x)$ of the system T there is a primitive recursive function $p_f(x, y)$ with the property*

$$f(x) = y \to th(p_f(x, y)) = \{f(x) = y\}.$$

Proof. By induction on the construction of $f(x)$. We consider the initial functions first, and look at the zero function $o(a)$ which satisfies $o(a) = 0$. $p_0(x, y)$ is given by

$$p_0(x, 0) = 6w(nu(x), w(0, 0)) + 1$$
$$p_0(x, y + 1) = 0,$$

(the number of the axiom $o(a) = 0$ is 0). Hence by the definition of $th(t)$

$$\begin{aligned} th(p_0(x, 0)) &= se(nu(x), 0, [o(a) = 0]) \\ &= se(nu(x), 0, se(nu(0), 1, [o(a) = b])) \\ &= \{o(x) = 0\} \end{aligned}$$

and the result follows. Similar constructions can be made for the other initial functions $s(a) = a + 1$, $i(a) = a$, $j(a, b) = b$ and $k(a, b, c) = c$.

We consider now composition, suppose $h(x) = f(g(x))$ and $p_f(x, y)$ and $p_g(x, y)$ satisfy

$$\begin{aligned} f(x) = y &\to th(p_f(x, y)) = \{f(x) = y\} \\ g(z) = x &\to th(p_g(z, x)) = \{g(z) = x\}. \end{aligned} \qquad \text{(i)}$$

As

$$g(z) = x \,\&\, f(g(z)) = y \to f(x) = y$$

we have

$$g(z) = x \,\&\, f(g(z)) = y \to th(p_f(x, y)) = \{f(x) = y\}. \qquad \text{(ii)}$$

We also have

$$g(z) = x \,\&\, f(x) = y \to f(g(z)) = y$$

so we get, by a simple extension of lemma 1, a primitive recursive function $W(t, u)$ satisfying

$$th(t) = \{g(z) = x\} \,\&\, th(u) = \{f(x) = y\} \to th(W(t, u)) = \{f(g(z)) = y\}$$

therefore by (i) and (ii)

$$g(z) = x \,\&\, f(g(z)) = y \to th(W(p_g(z, x), p_f(x, y))) = \{f(g(z)) = y\}.$$

In exactly similar manner we can construct W' such that

$$th(t) = \{f(g(z)) = y\} \to th(W'(z, t)) = \{h(z) = y\}$$

and combining these we get

$$h(z) = y \to th(W'(z, W(p_g(z, g(z)), p_f(g(z), y)))) = \{h(z) = y\}.$$

Hence $p_h(x, y)$ is defined by

$$p_h(x, y) = W'(x, W(p_g(x, g(x)), p_f(g(x), y))).$$

It is clear that parameters can be inserted into $f(x)$ and $g(x)$ without affecting the argument.

Finally we consider recursion, suppose $h(a)$ is given by

$$h(0) = 0$$
$$h(a+1) = g(h(a)).$$

$p_h(0, y)$ is defined as before and $p_h(x+1, y)$ is defined from $p_h(x, y)$ using the composition case above. For as $g(x)$ is a given function we have

$$g(x) = y \to th(p_g(x, y)) = \{g(x) = y\}$$

and by hypothesis

$$h(z) = x \to th(p_h(z, x)) = \{h(z) = x\}$$

so, defining W'' in an analogous way to W' using the equation $h(a+1) = g(h(a))$, we get

$$p_h(x+1, y) = W''(x, W(p_h(x, h(x)), p_g(h(x), y))),$$

hence in this case also $p_h(x, y)$ is primitive recursive. Again parameters can be inserted into $g(a)$ and $h(a)$ without affecting the argument and so lemma 2 follows.

Lemma 3 (GÖDEL's first incompleteness theorem). *If* $N = [se(nu(a), a) \mp th(z)]$ *then* $se(nu(N), N) = [se(nu(N), N) \mp th(z)]$, *and* $se(nu(N), N) \mp th(z)$ *is undecidable in the system* T, *if* T *is consistent.*

For the proof see [4].

Theorem 4 (GÖDEL's second incompleteness theorem).

$$th(y) \mp e(th(z)),$$

that is the statement of consistency for the system T, *is not decidable in* T.

Proof. As the substitution of a numeral for a variable is a valid deductive procedure there is, by lemma 1, a primitive recursive function $Y(x, t)$ with the property

(i) $$th(Y(x, t)) \mp \{f(x) = 0\} \to th(t) \mp [f(z) = 0].$$

Now let $G(z) = |se(nu(N), N), th(z)|$ and we have

$$\begin{aligned}
th(z) = se(nu(N), N) & \\
\to G(z) = 0 & \\
\to th(p_G(z, 0)) = \{G(z) = 0\} & \qquad \text{by lemma 2} \\
\to th(V(p_G(z, 0))) = \{se(nu(N), N) = th(z)\} & \qquad \text{by lemma 1} \\
\to e(th(V(p_G(z, 0)))) = \{se(nu(N), N) \mp th(z)\} & \\
\to th(Y(z, z)) \mp \{se(nu(N), N) \mp th(z)\} & \qquad \text{by hypothesis} \\
\to th(z) \mp [se(nu(N), N) \mp th(z)] & \qquad \text{by (i)} \\
\to th(z) \mp se(nu(N), N) & \qquad \text{by lemma 3,}
\end{aligned}$$

the result follows by reduction ad absurdum and lemma 3.

Theorem 5 (LÖB's theorem). *If* $th(x) = \{S(y)\} \to S(y)$ *is a theorem of* T *then so is* $S(y)$.

Proof. Let $[th(c) = se(a, a) \to S(b)] = f$ then

$$[th(c) = se(f, f) \to S(b)] = se(f, f),$$

so if $R(x, y) = 0$ is defined to be $th(x) = se(f, f) \to S(y)$ then

(a) $$R(x, y) = 0 \equiv th(x) = [R(c, b) = 0] \to S(y).$$

(b) If $R(x, y) = 0$ is derivable in T then so is $S(y)$.

For if $\vdash_T R(x, y) = 0$ we can find a number M, the number of this derivation, such that $\vdash_T th(M) = [R(c, b) = 0]$ and $S(y)$ follows by substitution and modus ponens using (a).

(c) $R(x, y) = 0$ is derivable in T.

By lemma 2 with $th(x) = [R(c, b) = 0]$ for $f(x) = y$ we have

$$th(x) = [R(c, b) = 0] \to th(q(x)) = \{th(x) = [R(c, b) = 0]\}$$

where $q(x) = p_{th}(x, [R(c, b) = 0])$, and by lemma 1 and (a) there is a primitive recursive function $V(t, u)$ with the property

$$th(t) = \{th(x) = [R(c, b) = 0]\} \,\&\, th(u) = \{R(x, y) = 0\} \to th(V(t, u)) = \{S(y)\}$$

so

$$th(x) = [R(c, b) = 0] \,\&\, th(u) = \{R(x, y) = 0\} \to th(V(q(x), u)) = \{S(y)\}.$$

But there is a second primitive recursive function $Y(x)$ with the property

$$th(x) = [R(c, b) = 0] \to th(Y(x)) = \{R(x, y) = 0\}$$

so we have

$$th(x) = [R(c, b) = 0] \to th(V(q(x), Y(x))) = \{S(y)\},$$

this and the hypothesis of the theorem give

$$th(x) = [R(c, b) = 0] \to S(y)$$

that is by (a)

$$R(x, y) = 0.$$

Theorem 5 now follows from (b) and (c).

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